RAJALAKSHMI ENGINEERING COLLEGE

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B2

Started on	Thursday, 13 August 2026, 11:34 AM
State	Finished
Completed on	Thursday, 20 August 2026, 10:53 AM
Time taken	6 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
#include<stdio.h>

int findfloor(int arr[], int low, int high, int x){
    if (low > high){
        return high;
    }
    int mid = (low + high ) / 2;

    if (arr[mid] == x){
        return mid;
    }

    if (arr[mid] > x){
        return findfloor(arr, low, mid - 1, x);
    }
    return findfloor(arr, mid + 1, high, x);
}
int main(){
    int m;
    scanf("%d ",&m);

    int arr[m];
    for (int i = 0; i < m; i++){
        scanf("%d",&arr[i]);
    }

    int x;
    scanf("%d",&x);

    int findindex = findfloor(arr, 0, m - 1, x);

    if(findindex >= 0){
        printf("%d\n",arr[findindex]);
    }
    else{
        printf("-1\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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